

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
iris = load_iris()
X = iris.data
y = iris.target
scaler = StandardScaler()
X = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
svm = SVC(kernel='linear')
svm.fit(X_train, y_train)
svm_pred = svm.predict(X_test)
print("SVM Accuracy:", accuracy_score(y_test, svm_pred))
print("SVM Confusion Matrix:")
print(confusion_matrix(y_test, svm_pred))
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
rf_pred = rf.predict(X_test)
print("\nRandom Forest Accuracy:",
      accuracy_score(y_test, rf_pred))
print("Random Forest Confusion Matrix:")
print(confusion_matrix(y_test, rf_pred))
plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
sns.heatmap(
    confusion_matrix(y_test, svm_pred),
    annot=True,
    fmt='d'
)
plt.title("SVM Confusion Matrix")

plt.subplot(1, 2, 2)
sns.heatmap(
    confusion_matrix(y_test, rf_pred),
    annot=True,
    fmt='d'
)
plt.title("Random Forest Confusion Matrix")

plt.tight_layout()
plt.show()
```

SVM Accuracy: 0.9777777777777777

SVM Confusion Matrix:

```
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
```

Random Forest Accuracy: 1.0

Random Forest Confusion Matrix:

```
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

